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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Customer Relationship Management (CRM) System for Small Businesses

Submitted by

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Academic Year: **2025 – 2026**

APRIL 2026

Semester: **I**

DECLARATION

I hereby declare that the project entitled "**PROJECT TITLE"** submitted in partial fulfillment of the requirements for the course **[Course Name]** is a **record of my original work** carried out under the guidance of **[Trainer Name]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"PROJECT TITLE"** is a **bona fide work** carried out by **[YOUR NAME] [Class/Batch]**, [**Department Name]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[Course Name]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

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Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

Abstraction in a Customer Relationship Management (CRM) system for small businesses refers to the process of simplifying the system by showing only the essential features to the user while hiding the complex technical details working behind the scenes. In such a CRM system, business owners and employees can easily manage customer information, track sales, communicate with clients, and generate reports without needing to understand how the system stores data, processes requests, or performs calculations. For example, when a user adds a new customer, they simply enter details like name, phone number, and email, while the system internally handles database storage, data organization, and security. Similarly, features like sales tracking and report generation present clear and simple outputs such as progress charts or summaries, even though they involve complex data processing in the background. This abstraction makes the CRM system user-friendly, especially for small business owners who may not have technical knowledge, and helps them focus on their core activities like improving customer relationships and increasing sales. By reducing complexity and improving usability, abstraction plays a key role in making CRM systems efficient, accessible, and practical for everyday business use.

1. INTRODUCTION

A Customer Relationship Management (CRM) system is a powerful tool that helps small businesses manage their interactions with customers in an organized and efficient way. In today's competitive market, maintaining strong relationships with customers is very important

for business growth and success. A CRM system allows businesses to store customer information such as contact details, purchase history, preferences, and communication records in one central place. This makes it easier for business owners and employees to understand customer needs, provide better service, and build long-term relationships. For small businesses, which often have limited resources and staff, a CRM system helps save time by automating tasks like sending follow-up messages, tracking sales, and managing leads. It also improves communication within the team by keeping everyone updated with the latest customer information. By using a CRM system, small businesses can increase customer satisfaction, improve sales performance, and make better decisions based on data. Overall, a CRM system plays a key role in helping small businesses grow, compete with larger companies, and maintain strong connections with their customers.

1.1 Objective

An objective is a clear and specific goal that a person or organization wants to achieve.
It gives direction and purpose to activities and actions.
Objectives help in planning and decision-making processes.
They are usually measurable and time-bound.
Objectives guide individuals to focus on desired outcomes.
They help in evaluating performance and progress.
In business, objectives support growth and development.
They ensure proper use of resources and efforts.
Objectives can be short-term or long-term based on needs.
Overall, an objective is a target that helps in achieving success systematically.

1. To manage and organize customer information effectively.
2. To improve customer satisfaction and service quality.
3. To increase sales and business opportunities.
4. To build strong and long-lasting customer relationships.
5. To automate tasks and improve business efficiency

1.2 Overview

Customer Relationship Management (CRM) is a system used by businesses to manage their interactions with customers. It helps small businesses store customer information in an organized way. CRM systems track customer details like contact information, purchase history, and preferences. They improve communication between the business and its customers. CRM also helps in managing sales, marketing, and customer service activities. It allows businesses to understand customer needs better. By using CRM, small businesses can build strong relationships with their customers. It helps increase customer satisfaction and loyalty. CRM systems can be manual or software-based. Overall, CRM helps small businesses grow by improving customer management and business efficiency.

1. CRM helps store and manage customer data efficiently.
2. It improves communication with customers.
3. It supports sales and marketing activities.
4. It increases customer satisfaction and loyalty.
5. It helps small businesses grow and make better decisions

2. PROGRAM ANALYSIS

2.1Problem Definition

Program analysis of a CRM system for small businesses focuses on understanding how the system works, what features it provides, and how it helps manage customer relationships effectively. The CRM program is designed to store and organize customer data such as contact details, communication history, and purchase records in a centralized database. It analyzes customer behavior and interactions, helping businesses identify patterns, improve customer service, and increase sales. The system also automates tasks like sending emails, reminders, and follow-ups, reducing manual work. By analyzing data through reports and dashboards, small businesses can make better decisions, improve marketing strategies, and build stronger customer relationships. Overall, CRM program analysis ensures that the system meets business needs, improves efficiency, and supports business growth.

2.2Problem Overview

Customer Relationship Management (CRM) is a system used by businesses to manage their interactions with customers. It helps small businesses store customer information in an

organized way. CRM systems track customer details like contact information, purchase history, and preferences. They improve communication between the business and its customers. CRM also helps in managing sales, marketing, and customer service activities. It allows businesses to understand customer needs better. By using CRM, small businesses can build strong relationships with their customers. It helps increase customer satisfaction and loyalty. CRM systems can be manual or software-based. Overall, CRM helps small businesses grow by improving customer management and business efficiency.

CRM helps store and manage customer data efficiently.
It improves communication with customers.
It supports sales and marketing activities.
It increases customer satisfaction and loyalty.
It helps small businesses grow and make better decisions.

2.3 Scopes and Limitations

Scope

The scope of this CRM system is to help small businesses manage customer information in a simple and efficient way. It allows users to store customer details such as ID, name, phone number, and email in a centralized system. The system supports basic operations like adding, viewing, updating, and deleting customer records. It improves customer communication, helps in maintaining organized data, and reduces manual paperwork. The system is easy to use, cost-effective, and suitable for small-scale businesses with limited resources. It can also be expanded in the future by adding features like sales tracking, reporting, and customer feedback.

The CRM system has some limitations as it is a basic version. It does not support advanced features like automation, data analytics, or integration with other business tools. The system stores data only temporarily (if not connected to a database), so information may be lost when the program is closed. It is designed for small businesses, so it may not handle large amounts of data efficiently. Security features are also limited, which may not fully protect sensitive customer information. Additionally, it may require technical knowledge to modify or upgrade the system.

3. SYSTEM DESIGN

The system design of a Customer Relationship Management (CRM) system for small businesses focuses on simplicity and efficiency. It includes a central database to store customer details like names, contacts, and purchase history. The design has key modules such as sales tracking, marketing, and customer support. It is often cloud-based, allowing access from different devices. The system also provides an easy-to-use interface for users. Basic security features are included to protect data. Overall, the design helps small businesses manage customer information smoothly and improve their operations

1. Customer Data Management Module

Stores and manages customer details like names, contacts, and purchase history

2. Sales Tracking Module

Helps monitor sales activities, orders, and performance

3. Marketing Module

Supports campaigns, promotions, and customer engagement.

4. Customer Support Module

Handles customer queries, complaints, and service requests.

5.User Interface Module

Provides an easy and user-friendly way to access the system.

4. METHODOLOGY

The methodology used for developing a Customer Relationship Management (CRM) system for small businesses follows a simple and structured approach. First, the requirements are collected by understanding the needs of the business and its customers. Then, the system is designed with modules like customer management, sales tracking, and support services. After the design phase, the system is developed using suitable technologies and tools. Next, testing is done to ensure the system works correctly without errors. Once tested, the system is implemented and used by the business. Finally, regular maintenance and updates are carried out to improve performance and meet changing business needs.

The development process includes the following stages:

- **Problem Analysis**
- **System Design**
- **Program Development**
- **Testing and Validation**
- **Result and Documentation**

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of restaurant operations.

The main problems identified in manual billing systems are:

• Manual calculations may lead to billing errors.
• Preparing bills manually takes more time.
• Difficult to handle multiple customer orders efficiently.
• Lack of automated records for transactions.

Based on these problems, the system requirements were defined to develop an automated billing application that simplifies restaurant billing operations.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

• Menu Display Module
• Order Selection Module
• Bill Calculation Module
• Invoice Generation Module
• Exit Module

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

After designing the system, the program is implemented using the **Java programming language**.

The following steps were followed during development:

• Creation of classes and methods
• Implementation of menu-driven logic
• Writing code for bill calculation
• Handling user input
• Displaying the final bill

Core Java concepts such as **classes, objects, loops, conditional statements, and methods** were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	**Expected Result**
Add a new customer with valid details	Customer information is successfully saved in the database
Add a customer with missing required fields	System shows an error message and does not save the data
Search for an existing customer	Correct customer details are displayed
Update customer information	Changes are saved and updated details are shown

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

The Customer Relationship Management (CRM) system for small businesses successfully improves the management of customer data and business activities. It helps in organizing customer information, tracking sales, and handling customer support efficiently. The system reduces manual work and saves time by automating tasks. It also improves communication with customers, leading to better satisfaction and loyalty. Overall, the CRM system increases productivity and supports the growth of small businesses.

The documentation of the CRM system includes detailed information about system design, modules, and functionalities. It provides user guides to help users understand how to operate

the system easily. Technical documentation explains the system architecture, database structure, and technologies used. It also includes instructions for installation, usage, and maintenance. Proper documentation ensures that the system can be used, updated, and managed effectively in the future.

4.6. Algorithm

1.Start
2.User logs into the CRM system using valid credentials
3.Display main menu with options (Customer, Sales, Marketing, Support)
4.User selects an option
5.If Customer Module selected:
  *Add / View / Update / Delete customer details
6.If Sales Module selected:
 *Create and manage orders, track sales
7.If Marketing Module selected:
 *Manage campaigns and promotions
8.If Support Module selected:
 *Handle customer queries and service requests
9.Store all data in the central database
10.Generate reports if requested
11.Logout from the system
12.End

4.7. Flowchart

Start
 ↓
Login
 ↓
Work (Customer / Sales / Support)
 ↓
Save
 ↓
Logout
 ↓
End

5. SOURCE CODE

```
import java.util.ArrayList;
import java.util.Scanner;

class Customer {
    String id, name, phone, email;

    Customer(String id, String name, String phone, String email) {
        this.id = id;
        this.name = name;
        this.phone = phone;
        this.email = email;
    }

    void display() {
        System.out.println("ID: " + id + ", Name: " + name +
                ", Phone: " + phone + ", Email: " + email);
    }
}

public class CRMSystem {
    static ArrayList<Customer> customers = new ArrayList<>();
    static Scanner sc = new Scanner(System.in);

    public static void addCustomer() {
        System.out.print("Enter ID: ");
        String id = sc.nextLine();
        System.out.print("Enter Name: ");
```

```
        String name = sc.nextLine();
        System.out.print("Enter Phone: ");
        String phone = sc.nextLine();
        System.out.print("Enter Email: ");
        String email = sc.nextLine();

        customers.add(new Customer(id, name, phone, email));
        System.out.println("Customer added successfully!\n");
    }

    public static void viewCustomers() {
        if (customers.isEmpty()) {
            System.out.println("No customers found.\n");
        } else {
            for (Customer c : customers) {
                c.display();
            }
            System.out.println();
        }
    }

    public static void updateCustomer() {
        System.out.print("Enter Customer ID to update: ");
        String id = sc.nextLine();

        for (Customer c : customers) {
            if (c.id.equals(id)) {
                System.out.print("Enter new Name: ");
                c.name = sc.nextLine();
                System.out.print("Enter new Phone: ");
                c.phone = sc.nextLine();
                System.out.print("Enter new Email: ");
                c.email = sc.nextLine();
                System.out.println("Customer updated successfully!\n");
                return;
            }
        }
        System.out.println("Customer not found.\n");
    }

    public static void deleteCustomer() {
        System.out.print("Enter Customer ID to delete: ");
        String id = sc.nextLine();
```

```
    for (Customer c : customers) {
        if (c.id.equals(id)) {
            customers.remove(c);
            System.out.println("Customer deleted successfully!\n");
            return;
        }
    }
    System.out.println("Customer not found.\n");
}

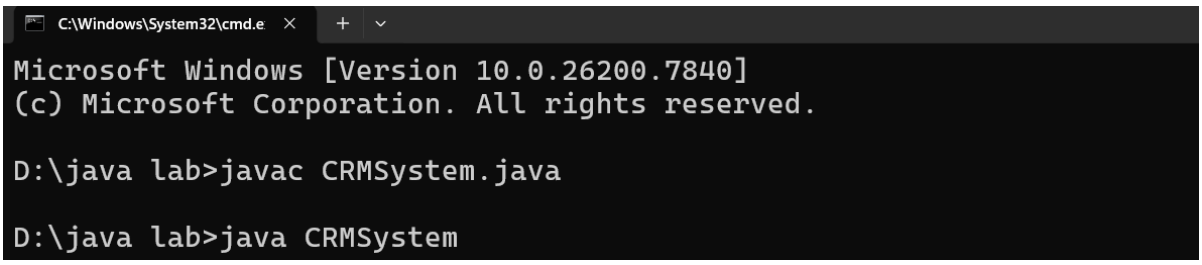
public static void main(String[] args) {
    while (true) {
        System.out.println("---- CRM MENU ----");
        System.out.println("1. Add Customer");
        System.out.println("2. View Customers");
        System.out.println("3. Update Customer");
        System.out.println("4. Delete Customer");
        System.out.println("5. Exit");

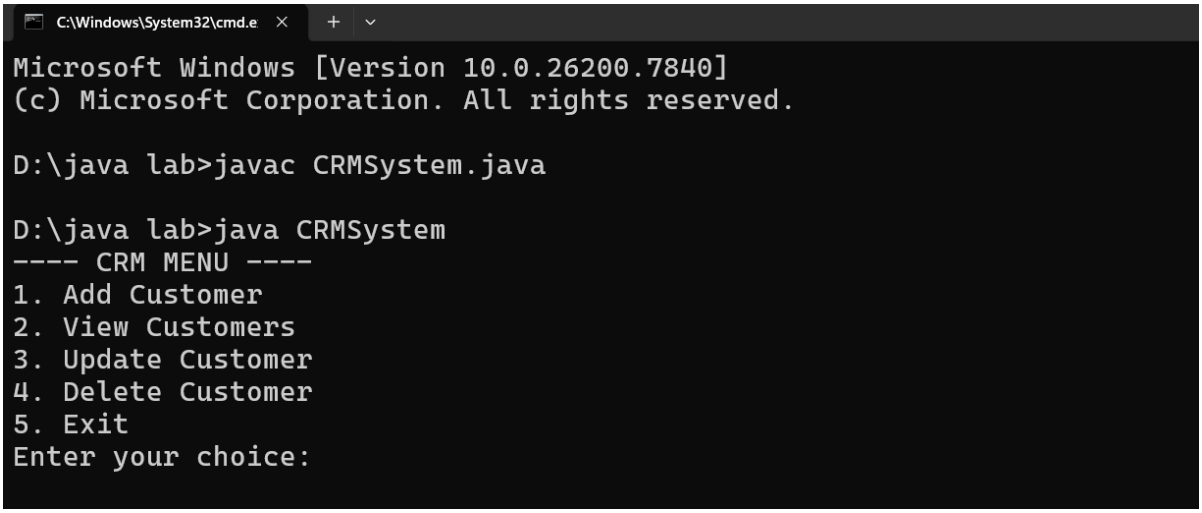
        System.out.print("Enter your choice: ");
        String choice = sc.nextLine();

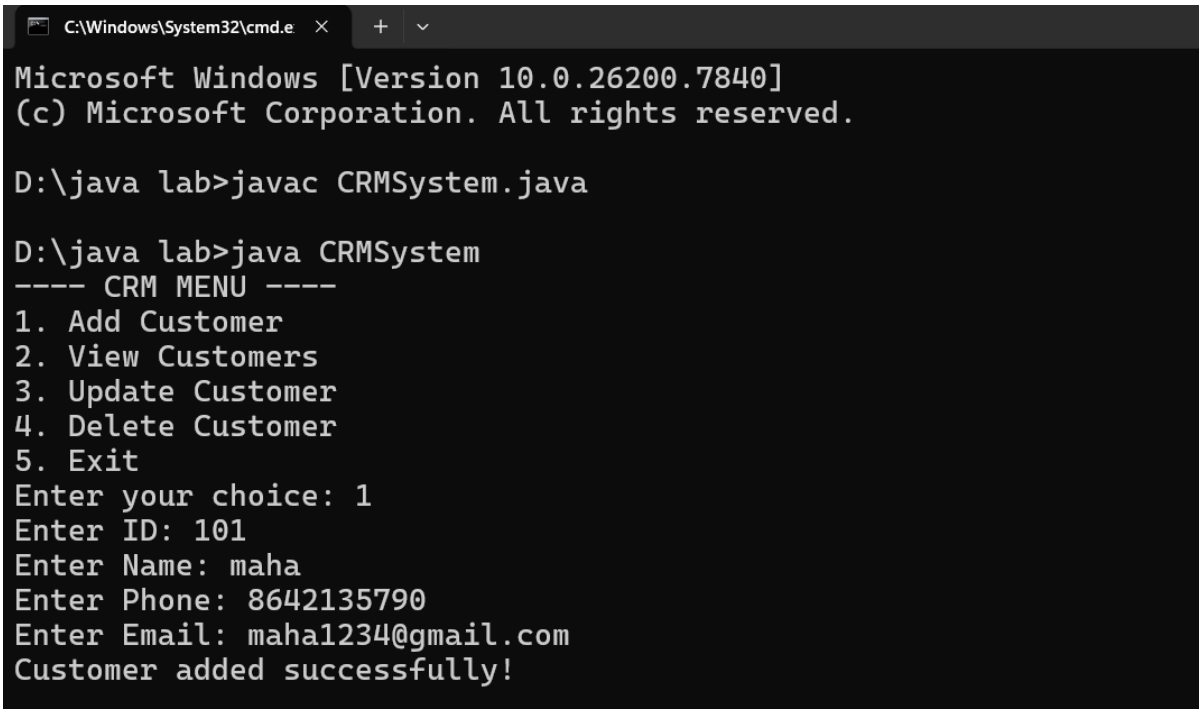
        switch (choice) {
            case "1":
                addCustomer();
                break;
            case "2":
                viewCustomers();
                break;
            case "3":
                updateCustomer();
                break;
            case "4":
                deleteCustomer();
                break;
            case "5":
                System.out.println("Exiting CRM System...");
                return;
            default:
                System.out.println("Invalid choice! Try again.\n");
        }
    }
```

```
    }
}
```

6. INPUT AND OUTPUT SCREENS




C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.
D:\java lab>javac CRMSystem.java
D:\java lab>java CRMSystem
---- CRM MENU ----
1. Add Customer
2. View Customers
3. Update Customer
4. Delete Customer
5. Exit
Enter your choice:


C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.
D:\java lab>javac CRMSystem.java
D:\java lab>java CRMSystem
---- CRM MENU ----
1. Add Customer
2. View Customers
3. Update Customer
4. Delete Customer
5. Exit
Enter your choice: 1
Enter ID: 101
Enter Name: maha
Enter Phone: 8642135790
Enter Email: maha1234@gmail.com
Customer added successfully!



7. DISCUSSION &CONCLUSION

The Customer Relationship Management (CRM) system helps small businesses manage customer information in an organized way. It improves communication between the business

and customers by keeping all details in one place. The system supports important activities like sales tracking, marketing, and customer service. It reduces manual work and saves time, making daily operations easier. However, small businesses may face challenges such as initial setup cost, need for basic training, and data security concerns. Despite these limitations, the CRM system is very useful for improving efficiency and customer satisfaction.

7.2 Conclusion

In conclusion, the CRM system is an effective tool for small businesses to manage customer relationships and business activities. It helps in increasing productivity, improving customer service, and supporting business growth. With proper use and maintenance, the system can provide long-term benefits and make business operations more smooth and successful

8. REFERENCES

The following resources were used during the development of the Customer Relationship Management(CRM)System for Small Business.

Books

Java:**The Complete Reference-Herbert schildt**

Websites

• **Oracle Java Documentation**- https://www.oracle.com/java
• **GeeksforGeeks**-https://www.geeksforgeeks.org
• **W3Schools Java Tutorial**-https://www.w3schools.com
• **Javapoint Java Tutorial**-https://www.javatpoint.com